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Dynamic Graph-based Deep Reinforcement Learning with Long and Short-term Relation Modeling for Portfolio Optimization

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Abstract

Portfolio optimization is a significant concern in finance. Existing research on portfolio optimization fails to adequately learn from the long and short-term relationships among equities, which inevitably leads to suboptimal performance. In this paper, we propose a Dynamic Graph-based Deep Reinforcement Learning (DGDRL) for optimal portfolio decisions. We achieve this goal by devising two mechanisms for *naturally* modeling the financial market. Firstly, we utilize the static and dynamic graphs to represent the long and short-term relations, which are then *naturally* represented by the proposed multi-channel graph attention neural network. Secondly, compared with the traditional two-phase approach, forecasting equity's trend and then weighting them by combinatorial optimization, we *naturally* optimize the portfolio decisions, which could directly guide the model to converge to optimal rewards. Through extensive experiments on three real-world datasets, we have demonstrated that our method significantly outperforms state-of-the-art benchmark methods in portfolio management. Furthermore, the evaluation of the industrial trading system has shown the applicability of our model to real-world financial markets.

CCS Concepts

• **Applied computing** → **Economics**; • **Information systems** → **Data mining**.

Keywords

Graph Neural Networks, Deep Reinforcement Learning, Portfolio Optimization

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1 Introduction

Effective portfolio strategies can enhance the stability of investors' wealth by seeking high returns while mitigating risks [19]. Recently, modern portfolio theory has acknowledged the increasing importance of the relationships between assets [6]. Specifically, it is necessary to understand how assets influence each other's and the market's price movements. For example, Hendricks *et al.* [13] found that assets' relationships in industry and supply chain affect their long-term stock price performance and equity risk. Meanwhile, as a dynamic system, the stock market exhibits complex short-term temporal dependencies [26]. Figure 1 displays a typical example of long-term and short-term relations between assets. In the blue part, Microsoft, Apple, Activision and Electronic Arts are interconnected as they all belong to the Internet sector, with Activision and Electronic Arts having a stronger relation due to their shared involvement in the entertainment concept. Relations established based on explanatory theories such as sector and concept are long-term, reflecting the underlying supply chains and fundamental information of assets. On the other hand, relations arising from market fluctuations resulting from investors' gaming behavior are short-term. For instance, in the orange part, during the months of February and March 2021, the market volatility caused Microsoft and Electronic Arts' stock prices to exhibit similar variation trends, suggesting a relation between the two assets. Although such short-term relationships currently still remain difficult to model accurately, they can be regarded as hidden patterns deserving of further investigation [38]. Both the long and short-term relations jointly conduct a synthetical impact on the ever-changing financial markets [12, 33, 37].

Conventional research on modeling the long and short-term relations mainly leverages the temporal features [1, 42], which inevitably dismisses stocks' inter-connected relations. For example,

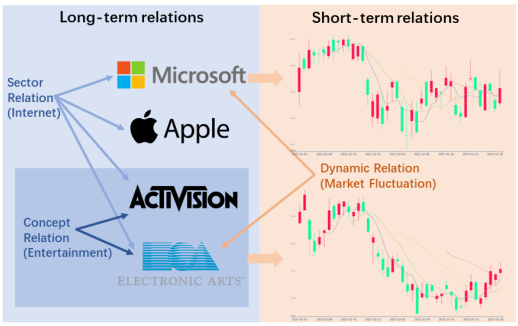


Figure 1: A typical example of relations between assets. The blue section represents static long-term relations built by sectors and concepts, while the orange section represents dynamic short-term relations formed by market fluctuations.

TTIO [20] uses mathematical summarization of historical price and volume data as technical indicators to model long and short-term price relations. Lately, graph neural networks have been widely employed to naturally model equity relations and achieve significant progress [9, 36]. For instance, THGNN [36] utilizes temporal and heterogeneous graphs to learn the relations among price movements in financial time series. However, existing graph-based models, typically by forecasting equity's prices first and then weighting them by combinatorial optimization, cannot directly infer portfolio weights which may lead to suboptimal performance [44].

Recently, several methods based on deep reinforcement learning (DRL) framework are then introduced to naturally optimize the portfolio weights, offering advantages in pursuing directed rewards [8, 23, 40]. For example, DeepTrader [35] employs a DRL-based portfolio optimization to tackle the risk-return balancing problem by embedding macro market conditions as an indicator. However, most DRL-based portfolio optimization methods [22, 39, 43] mainly focus on utilizing historical technical data for optimization, which does not include their graph-based relations. Only a few pioneers [29, 30] employ graph convolution reinforcement learning(RL) for portfolio management which only supports static relations. As is well-known, incorporating short-term dynamic relations is crucial for portfolio modeling [45]. Compared to static relation modeling, dynamically constructing stock relations poses significant challenges on subsequent integration and utilization of information. Because the transition from static graphs to dynamic graphs substantially increases the complexity of the optimal policy search, requiring agents to allocate more time to action value exploration.

Therefore, in this paper, we propose a novel **D**ynamic **G**raph-based **D**eep **R**einforcement **L**earning (DGDRL) framework for portfolio optimization. Our method could *naturally* model the long and short-term stock relations in dynamic graphs and *naturally* optimize the portfolio performance in the DRL framework. In particular, we first construct stocks' relations into dynamic (short-term relation) and static (long-term relation) graphs. Then, we devise a multi-channel graph attention layer for dynamic representation learning and directly inferring the portfolio weights as outputs. The actions are optimized through the DRL framework to converge to optimal rewards. The extensive experiments on three real-world datasets demonstrate the superior performance of our proposed method. In conclusion, our contributions can be summarized as:

- We propose a novel dynamic graph-based deep reinforcement learning framework for portfolio optimization, which could *naturally* model the long and short-term relations of financial equities and naturally optimize portfolio decisions.
- We construct dynamic graphs to capture the relationships among stocks, and propose a novel multi-channel graph attention layer for learning dynamic representations of graphs. Then we directly optimize the agent outputs by the deep reinforcement learning framework, which could effectively guides the model to converge to optimal rewards.
- Extensive experiments are conducted on three real-world datasets to demonstrate the superiority of DGDRL over state-of-the-art baselines. Additionally, We validated the effectiveness of our proposed method in a real-world trading system. The system was provided by Emoney Inc., a leading financial service provider in China.

2 Methodology

2.1 Model Architecture

In this section, we propose the DGDRL method as illlustrated in Figure 2. Our method primarily consists of three components: (a) the construction of an asset relationship graphs, including static and dynamic graphs; (b) dynamic multi-channel graph attention networks incorporating a Long Short-Term Memory [14] with History state Attention (LSTM-HA) and a novel Multi-Channel Graph Attention Network (MGAN) layers; and (c) interactive training based on the RL framework, employing actor-critic policy optimization.

2.2 Multi-channel Graph Construction

To effectively leverage the relational information among assets, we model both the dynamic graph (short-term relations) and the static graph (long-term relations) separately, thereby accommodating joint learning of the graph structure and the DRL framework. We will first introduce the process of constructing the graph.

Temporal Graph Neighborhood. The neighborhood of a node v in graph g is defined as $\mathcal{N}_g(v) = \{u_i \mid d_{sp}(v, u_i) \leq d_{\mathcal{N}}, |t_v - t_{u_i}| \leq t_{\mathcal{N}}\}$, where $d_{sp}(\cdot|\cdot)$ denotes the shortest path length between nodes, the $d_{\mathcal{N}}$ and $t_{\mathcal{N}}$ denote the distance threshold for the nodes in spatial and temporal dimensions. And the $\mathcal{N}_g(\cdot)$ is the neighbor function of graph $g \in \{d, s\}$, where d means dynamic and s means static.

Static Graph Construction. Our static graph is constructed using NLP techniques to analyze the fundamental information and business descriptions of assets. Specifically, we employed two pre-trained text analysis models from Hugging Face to assess the sector and concept relevance for each stock asset based on their fundamental information and business descriptions. Subsequently, the scores are calculated for correlation using the Pearson correlation coefficient, which served as the weight for the relations between the stocks. We perform threshold screening on the graph in order to increase the signal-to-noise ratio. The relationship between two assets is deemed to have no significant impact if their shortest distance exceeds the threshold, in which case they are not connected in the static graph; otherwise, edges are present between them.

Dynamic Graph Construction. In contrast to static graphs, dynamic graphs are based solely on market trend data, such as open, high, low and close. We utilize the Spearman's Rank Correlation

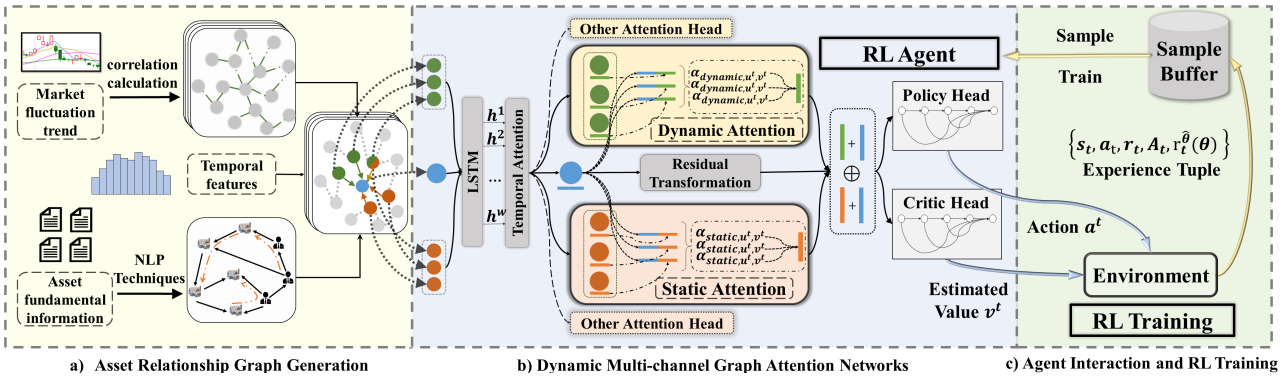


Figure 2: The proposed DGDRL framework for portfolio management. The first part is the construction of multi-channel graphs, which construct heterogeneous long- and short-term relation graphs of stocks. The second part consists of dynamic multi-channel graph attention networks, which can naturally perform dynamic representation learning by combining information from long and short-term neighbors. The third part is the agent interaction and RL training, where a policy head and a critic head engage in environment-based interaction and directly infer portfolio weights as outputs.

Coefficient to calculate the rank similarity of stock returns within a given time window, thereby facilitating a more robust exploration of the nonlinear dynamic relations among stocks. The stock correlation can be expressed as $\rho = \frac{1}{l_w}\sum_{t=1}^{l_w}(1 - \frac{6\sum_{i=1}^{i=N}(d_i^t)^2}{N\cdot(N^2-1)})$, where l_w denotes the length of the look-back window, and d_i^t represents the difference in rank values of the corresponding feature i between two stocks on day t. The existence of edges is determined using the value of correlation coefficient, similar to the threshold screening process for the static graph.

Information aggregation. The input of our model comprises the temporal sequential features of the stocks, as well as the static graph and dynamic graphs described above. These three types of information are aggregated into complete dynamic multi-channel graphs, denoted as $G = (V, \{E_{Dynamic}, E_{Static}\})$. Vertices represent the information of individual stocks, and edges represent two types of relations between stocks: the full-cycle and static asset relation and the temporal and dynamic asset relations. As depicted in the Figure 2 (a), each individual stock simultaneously exists in both the static and dynamic graphs, being influenced by the long-term and short-term relationships with other stocks.

2.3 Dynamic Multi-channel Graph Attention

The initial representations of stocks in day t consist of historical information from trading day $t - l_w$ to trading day t. The historical states can be represented by a sequence $\mathbf{X}_{in}^t = \{\mathbf{X}^1, \ldots, \mathbf{X}^k, \ldots, \mathbf{X}^{l_w}\}$. We employ LSTM recursively to extract hidden representations at each time step as $\mathbf{H}_k^t = \text{LSTM}(\mathbf{H}_{k-1}^t, \mathbf{X}^k), k \in [1, l_w]$, accompanied by a historical state attention mechanism to unearth the significance of intermediate moment $\mathbf{H}_k^t$, thereby mitigating information loss. Finally, the representation $\mathbf{E}_l^t \in \mathbb{R}^{N\times d_{hid}}$ in day t is $\mathbf{E}_l^t = \sum_{k=1}^{l_w}\alpha_k\mathbf{H}_{l_w}^t$, and α_k denotes the attention weight between $\mathbf{H}_k^t$ and $\mathbf{H}_{l_w}^t$:

$$e_i = \tanh([\mathbf{H}_k^t, \mathbf{H}_{l_w}^t]\mathbf{W}_1 + \mathbf{X}^{l_w}\mathbf{W}_2)\boldsymbol{V}_e^{\mathrm{T}}, \alpha_k = \frac{\exp(e_k)}{\sum_{i=1}^{l_w}\exp(e_i)}, \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{2d_{hid}\times d_{hid}}, \mathbf{W}_2 \in \mathbb{R}^{d_{in}\times d_{hid}}$ are both the projection weights, and $\boldsymbol{V}_e \in \mathbb{R}^{d_{hid}}$ is the weight vector.

Then we propose a multi-channel graph attention mechanism on the heterogeneous inputs, leveraging the output $\mathbf{E}_l^t$ of the LSTM-HA and the multi-channel asset graph G. The MGAN layers can simultaneously aggregate long and short-term relations from the dynamic and static graphs. The information aggregation is formulated for each type of graph g as follows:

$$\mathbf{E}_g^t = \text{Concat}(\text{TgaHead}_g^1, \ldots, \text{TgaHead}_g^{n_{tga}})\mathbf{W}_{g,o}, \quad (2)$$

where $\mathbf{E}_g^t \in \mathbb{R}^{N\times d_{att}}$ represents the output of the multi-channel graph attention layer on day t, and $\mathbf{W}_{g,o} \in \mathbb{R}^{n_{tga}\times d_{hid}\times d_{att}}$ represents the attention output projection matrix, n_{tga} represents the number of attention heads, and each head $\text{TgaHead}_g^i \in \mathbb{R}^{N\times d_{hid}}$ of the MGAN layer is defined as follows:

$$\text{TgaHead}_g^i = \sum_{v^t\in V}\sigma\Big(\sum_{u^t\in\mathcal{N}_g(v^t)}\alpha_{u^t,v^t}^i\boldsymbol{e}_{u^t}\Big), \quad (3)$$

where σ represents the activation function, $\boldsymbol{e}_{u^t} \in \mathbb{R}^{d_{hid}}$ represents the temporal embedding of stock u in time t, and the α_{u^t,v^t}^i stands for the importance of relation edge (u^t, v^t) in i-th head, which is defined as follows:

$$\alpha_{u^t,v^t}^i = \frac{\exp(\text{LeakyRelu}(\boldsymbol{a}_{g,i}^{\mathrm{T}}[\boldsymbol{e}_{u^t} \parallel \boldsymbol{e}_{v^t}]))}{\sum_{k^t\in\mathcal{N}_g(v^t)}\exp(\text{LeakyRelu}(\boldsymbol{a}_{g,i}^{\mathrm{T}}[\boldsymbol{e}_{k^t} \parallel \boldsymbol{e}_{v^t}]))}, \quad (4)$$

where $\boldsymbol{a}_{g,i} \in \mathbb{R}^{2d_{hid}}$ denotes the trainable graph attention weight vector of the specific graph g and i-th head.

Subsequently, we employ an efficient method to aggregate heterogeneous information from multi-channel graphs and sequential embeddings. We define the information sources as three types of embedding, which are information from the original output of LSTM-HA $\mathbf{E}_{ori}^t$, the short-term neighbors in dynamic relation graph $\mathbf{E}_d^t$ and the long-term neighbors in static relation graph $\mathbf{E}_s^t$, respectively. $\mathbf{E}_{ori}^t \in \mathbb{R}^{N\times d_{att}}$ is derived from $\mathbf{E}_l^t$ through a residual transformation. It can be expressed as $\mathbf{E}_{ori}^t = \mathbf{E}_l^t\mathbf{W}_{ori} + \boldsymbol{b}_{ori}$, where $\mathbf{W}_{ori} \in \mathbb{R}^{d_{hid}\times d_{att}}$ and $\boldsymbol{b}_{ori} \in \mathbb{R}^{d_{att}}$ represent the trainable residual projection parameters. $\mathbf{E}_d^t$ and $\mathbf{E}_s^t$ are derived from the multi-channel graph attention mechanism. Subsequently, we can

obtain the ultimate representations of all stocks on day t as:

$$\mathbf{E}^t_{final} = \text{Concat}\left(\mathbf{E}^t_d + \mathbf{E}^t_{ori}, \mathbf{E}^t_s + \mathbf{E}^t_{ori}\right) \tag{5}$$

2.4 Optimization via Reinforcement Learning

The final representations of stocks are then forwarded to both the actor head and the critic head of the agent for individual predictions. And we propose a Residual MLP (RMLP) for the actor and critic to better leverage the aggregated information as follows:

$$\begin{aligned}
\mathbf{E}^t_{a_1} &= \text{Dropout}(\sigma(\mathbf{W}_{a_1}\mathbf{E}^t_{final} + \boldsymbol{b}_{a_1})),\\
\mathbf{E}^t_{a_{i+1}} &= \text{Dropout}(\sigma(\mathbf{W}_{a_{i+1}}\mathbf{E}^t_{a_i} + \boldsymbol{b}_{a_{i+1}})),\\
E^t_{score} &= \sigma(\text{Concat}(\mathbf{E}^t_{a_1}, \ldots, \mathbf{E}^t_{a_{n_{rmlp}}})\boldsymbol{W}_a + b_a),
\end{aligned} \tag{6}$$

where $\mathbf{W}_{a_1} \in \mathbb{R}^{2d_{att} \times d_{att}/n_{rmlp}}$, $\boldsymbol{b}_{a_1} \in \mathbb{R}^{d_{att}/n_{rmlp}}$, $b_a \in \mathbb{R}$,

$\mathbf{W}_{a_i} \in \mathbb{R}^{d_{att}/n_{rmlp} \times d_{att}/n_{rmlp}}$, $\boldsymbol{b}_{a_i} \in \mathbb{R}^{d_{att}/n_{rmlp}}$, $\boldsymbol{W}_a \in \mathbb{R}^{d_{att} \times 1}$, n_{rmlp} refers to the number of RMLP layers and σ denotes the activation function. The portfolio weight a^t_i for stock i in day t is calculated as $a^t_i = \frac{\exp(E^t_{score_i})}{\sum_{j=1}^{N}\exp(E^t_{score_j})}$ and the portfolio weight vector is $\boldsymbol{a}_t = \{a^t_1, \ldots, a^t_N\}$. The investment interaction over a time period T is converted into a state-action-reward trajectory τ of the agent, which is stored in a buffer for future sampling like $\{\mathbf{s}_1, \boldsymbol{a}_1, r_1, \ldots, \mathbf{s}_t, \boldsymbol{a}_t, r_t, \ldots, \mathbf{s}_T, \boldsymbol{a}_T, r_T\}$. The $\mathbf{s}_t$ represents the current state of assets in trading day t. The $\boldsymbol{a}_t$ is a N dimensional portfolio weight vector, and r_t denotes the portfolio reward after executing $\boldsymbol{a}_t$. Upon implementing $\boldsymbol{a}_t$ on day t, the environment calculates the investment return rate brought by $\boldsymbol{a}_t$ based on the real-time stock market information and returns it as a reward on day $t+1$. Subsequently, after collecting the trajectory, we conduct multiple samplings and utilize the sampled mini-batches to jointly train the actor and critic efficiently and enhance the training stability. The complete training algorithm is outlined algorithm 1.

Algorithm 1 DGDRL Training Algorithm

Input: Initial Actor parameters $\theta = \theta_{old}$, initial exploration Actor parameters π^0_{old}, initial Critic parameters μ, batch size L, re-sampling times M

for $k = 0, 1, 2, \ldots$ **do**
  collect the interaction trajectory τ by running trajectory strategy π^k_{old} for T time-steps
  Compute discounted cumulative rewards $G_1, \ldots, G_T$, where $G_t = \sum_{k=t}^{T}\gamma^{k-t}r_k$, γ is the discounted factor
  Compute advantage estimates $A_1, \ldots, A_T$, where $A_t = G_t - \text{Critic}(\mathbf{s}_t)$, Critic is the critic head network
  for $i = 1, 2, \ldots, len(\tau) * M/L$ **do**
    Sample the trajectory τ and gain $mini\text{-}batch_i$
    Compute the divergence $r^{\theta_{old}}_t(\theta) = \frac{\pi_\theta(\boldsymbol{a}_t|\mathbf{s}_t)}{\pi_{\theta_{old}}(\boldsymbol{a}_t|\mathbf{s}_t)}$
    Update Actor by optimizing $\arg\max_\theta \frac{1}{|L|}J^\theta_{\text{clip}}$, where $J^\theta_{\text{clip}} = \sum_{(\mathbf{s},\boldsymbol{a})\in\tau}\min(r^{\theta_{old}}_t(\theta)A_t, \text{clip}(r^{\theta_{old}}_t(\theta), 1-\epsilon, 1+\epsilon)A_t)$
    Update Critic by optimizing $\arg\min_\mu \frac{1}{|L|}J(\mu)$, where $J(\mu) = \frac{1}{|\tau|}\sum_{t=1}^{|\tau|}\text{MSE}(\text{Critic}(\mathbf{X}^t_{in}), G_t)^2$
  end for
  $\theta_{old} \leftarrow \theta$
end for

3 Experiments

3.1 Experimental Setup

Datasets. We have collected three datasets to ensure the comprehensiveness and reliability of the experiment. They are 50 stocks from the constituent stocks of Shanghai Stock Exchange (SSE50) Index in Chinese A-share market, 30 stocks from the constituent stocks of Dow Jones Industrial Average (DOW30) and 96 stocks from the constituent stocks of National Association of Securities Dealers Automated Quotations (NDX100) in the U.S.market. We used historical price and volume features from 2020 to 2022 as our training set , and data from 2023 as our test set. We filtered out a small subset of stocks listed after 2020 with incomplete information.

Baselines and Metrics. We compared our proposed methods with following baselines: Market, which represents the stock market index; BLSW [15] and CSM [27], two basic manual trading strategies; Transformer [31], TRA [21], CGM [45], THGNN [36], FactorVAE [7], and CTTS [41], state-of-the-art methods using DL framework; PPO [28], AlphaStock [34], DeepTrader [35] and DeepPocket [30], state-of-the-art methods in DRL framework. The metrics employed in our study are as follows: Annualized Rate of Return (ARR), Annualized Volatility (AVol), Maximum DrawDown (MDD), Annualized Sharpe Ratio (ASR), Calmar Ratio(CR) and Information Ratio (IR).

Experiment Details. We utilize the PyTorch library to implement the model and used the Adam optimizer for model training. The training process was run on a server with 32G memory and single NVIDIA Tesla V100 GPU. For the full training set, entire training process needs about 12 hours. In terms of parameters, we set the encoding dimension d_{hid} to 128, the number of attention heads n_{tga} to 8, d_{att} in graph attention layers to 512, the number of RMLP layers n_{rmlp} to 4, batch size L for the mini-batch to 128, the re-sampling times M to 16, and the clipping ratio ϵ to 0.25.

3.2 Performance Analysis

We performed backtesting experiments on three datasets to validate the investment performance of DGDRL compared to the baselines. The evaluation results are shown in Table 1. DGDRL has the best ARR, MDD, ASR, CR and IR on all three datasets, which clearly demonstrates the excellent performance of DGDRL on portfolio optimization. The high ARR demonstrates the strong profit-generating capability of DGDRL, while the low absolute MDD indicates that DGDRL avoids significant capital losses. The high ASR, CR, IR show that DGDRL has a decent balance between risk and reward and does not just pursue return or avoid risk, despite the fact that the values of Avol are not particularly good. In investment, it is unrealistic to achieve high returns while bearing very low risk, so it is necessary to accept a certain level of risk. Moreover, compared to existing state-of-the-art methods, with the exception of the comparison with low-yield models such as DeepPocket, DGDRL's Avol are not significantly inferior. The comparison of DeepPocket and DGDRL revealed that, DeepPocket is more stable and less risky, but it offers very low investment returns. In contrast, DGDRL achieves substantial increases in returns while effectively managing risks, which aligns more closely with investors' investment requirements and

Table 1: Overall performance of all models on three datasets. The best performance is highlighted in bold. ARR measures the portfolio return effect, where the higher is the better. AVol and MDD measure the investment risk where the lower absolute value is better. ASR, CR, and IR quantify the profit under a unit of risk, where the higher is the better.

Datasets	SSE50						DOW30						NDX100					
Models	ARR	AVol	MDD	ASR	CR	IR	ARR	AVol	MDD	ASR	CR	IR	ARR	AVol	MDD	ASR	CR	IR
Market	-0.122	**0.140**	-0.212	-0.876	-0.577	-0.885	0.137	0.114	-0.091	1.202	1.514	1.191	0.414	0.163	-0.105	2.538	3.956	2.215
BLSW	-0.210	0.247	-0.386	-0.852	-0.544	-0.869	0.310	0.208	-0.215	1.490	1.441	1.538	0.583	0.213	-0.111	2.734	5.225	2.331
CSM	0.012	0.241	-0.363	0.051	0.034	0.173	0.194	0.170	-0.124	1.142	1.567	1.141	0.389	0.232	-0.176	1.677	2.217	1.561
Transformer	0.005	0.231	-0.259	0.021	0.019	0.136	0.250	0.146	-0.098	1.712	2.540	1.608	0.552	0.196	-0.106	2.821	5.221	2.355
TRA	0.045	0.189	-0.184	0.237	0.243	0.327	0.307	0.154	-0.084	1.989	3.639	1.819	0.611	0.237	-0.099	2.576	6.118	2.139
CGM	0.081	0.194	-0.152	0.419	0.535	0.501	0.238	0.157	-0.094	1.517	2.546	1.445	0.592	0.188	-0.130	3.143	4.552	2.575
THGNN	0.046	0.200	-0.175	0.230	0.262	0.325	0.302	0.174	-0.100	1.713	3.020	1.609	0.616	0.180	-0.095	3.421	6.504	2.768
FactorVAE	0.034	0.173	-0.182	0.196	0.186	0.279	0.241	0.158	-0.103	1.526	2.346	1.452	0.477	0.196	-0.119	2.428	3.990	2.093
CTTS	0.062	0.200	-0.169	0.309	0.366	0.401	0.362	0.215	-0.121	1.684	2.983	1.551	0.665	0.189	-0.129	3.524	5.147	2.809
PPO	0.036	0.215	-0.177	0.166	0.202	0.271	0.384	0.151	-0.107	2.543	3.671	2.237	0.683	0.199	-0.128	3.440	5.349	2.734
AlphaStock	0.198	0.174	-0.163	1.134	1.209	1.126	0.510	0.178	-0.130	2.867	3.930	2.416	0.720	0.197	-0.091	3.661	7.939	2.870
DeepTrader	0.093	0.192	-0.116	0.483	0.804	0.560	0.362	0.151	-0.097	2.393	3.736	2.128	0.725	0.221	-0.127	3.282	5.701	2.591
DeepPocket	0.003	0.148	-0.168	0.019	0.017	0.092	0.161	**0.113**	-0.086	1.434	1.882	1.390	0.443	**0.161**	-0.098	2.749	4.516	2.366
DGDRL	**0.341**	0.196	**-0.085**	**1.743**	**4.012**	**1.604**	**0.697**	0.154	**-0.049**	**4.531**	**14.338**	**3.531**	**0.882**	0.180	**-0.076**	**4.892**	**11.651**	**3.616**

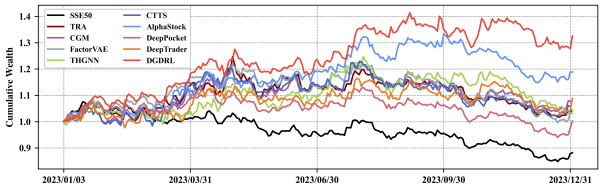


Figure 3: The accumulated returns gained in the test set (2023) of DGDRL, SSE50 index and primary baselines.

preferences. The difference can be attributed to that DeepPocket utilizes the static relationships of assets, leading to relatively fixed investment strategies, lacking dynamic factors used in DGDRL. Moreover, we can observe that methods that dynamically model the relationships between stocks, such as Alphastock exhibits relatively favorable investment performance. This, to some extent, also underscores the significance of incorporating dynamic relationships between stocks into the formulation of investment strategies. Furthermore, DRL-based methods demonstrate superior investment performance compared to DL methods on the whole, underscoring the efficacy of the DRL framework.

In order to further analyze the investment return of DGDRL over the test period, we report the cumulative return curve on SSE50 of DGDRL and main baselines in Figure 3. It can be observed that DGDRL nearly maintained its lead in returns over all baselines throughout the entire period. Even in situations where the overall market is in a downward trend with negative index returns, DGDRL demonstrates its adaptability to the market and the ability to achieve a stable and efficient long-term investment strategy.

3.3 Ablation Study

We investigated the impact of adjusting the graph structure and training architecture of DGDRL on model performance. We recorded four primary metrics for evaluation. DG-NG refers to DGDRL without utilizing graph structures. DG-ND represents DGDRL with the dynamic graphs removed, while DG-NS removes the static graph. DG-DL uses DL structure instead of RL training structure.

Table 2: Performance evaluation of ablated models for portfolio management on three datasets.

Dataset	Metric	DG-NG	DG-ND	DG-NS	DG-DL	DGDRL
SSE50	ARR	0.046	0.276	0.287	0.215	**0.341**
	ASR	0.253	1.501	1.448	1.340	**1.743**
	CR	0.244	2.853	2.738	1.400	**4.012**
	IR	0.340	1.423	1.377	1.299	**1.604**
DOW30	ARR	0.354	0.472	0.611	0.529	**0.697**
	ASR	2.398	3.187	3.989	3.209	**4.531**
	CR	4.109	6.597	10.297	4.343	**14.338**
	IR	2.136	2.697	3.205	2.671	**3.531**
NDX100	ARR	0.695	0.712	0.742	0.756	0.882
	ASR	3.772	3.984	4.016	4.055	**4.892**
	CR	5.945	9.554	6.460	7.593	**11.651**
	IR	2.970	3.113	3.112	3.128	**3.616**

Effectiveness of Dynamic Graphs and Static Graphs. As observed in Table 2, DGDRL outperforms DG-ND and DG-NS in all three markets because MGAN, which incorporates information from both dynamic and static graphs, models the relations between assets more efficiently over a long and short-term time span. As a result, MGAN can simultaneously capture short dependencies in dynamic graphs and long dependencies in static graphs effectively, which enables our model to place a higher weight on stocks that are more promising and to amass more return at the end of the whole trading period. It effectively demonstrates the efficiency of our information aggregation and utilization.

Effectiveness of RL Training Structure. As demonstrated in Table 2, it is evident that the DG-DL cannot achieve optimal performance across all datasets after transitioning the training structure from the RL to DL. This is due to the fact that DG-DL initially seeks to prognosticate the up and down of stocks prior to taking any action, with the two-stage sequence having a complex, nonlinear relationship with an investor's ultimate objective of maximizing return on investment. Clearly, replacing RL with DL for training



Figure 4: The interface of investment portfolio based on our proposed DGDRL method. We selected two stock's investment signals generated as an illustrative example.

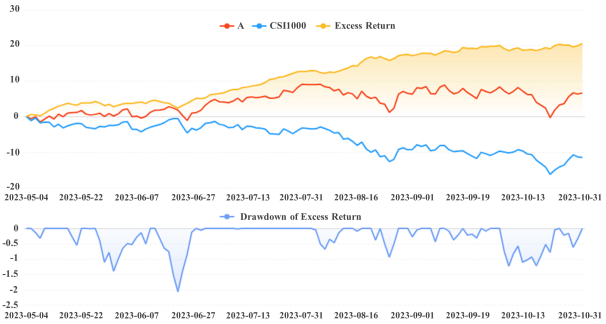


Figure 5: The observational evaluation results. The observation period is from 2023-05-01 to 2023-10-31.

results in performance that falls short of expectations. It elucidates that our model training can directly optimize investment returns.

3.4 Model Interpretability

We attempted to visualize the dynamic and static attention weights to investigate the learning process of DGDRL on dynamic graphs (short-term relations) and static graphs (long-term relations). For multiple attention heads, we averaged the weights, and the results are shown in Figure 6, where both the x and y-axes represent the 50 stocks in SSE50. Figure 6 (a) and (b) represent dynamic attention weights after one epoch of training and after full training, respectively. It can be observed that when training for just one epoch, the model does not adapt well to dynamic short-term changes, with imperfect information extraction from neighboring nodes, primarily focusing on self-information. However, after completing training, the model allocates different weights to neighbor nodes and self-information based on the characteristics of each stock. This indicates the dynamic instability of short-term relationships, requiring complex training for learning. Figure 6 (c) and (d) represent static attention weights after one epoch of training and after full training, respectively. It can be observed that, for both one epoch of training and full training, the model can focus on self and neighboring nodes' information, with minimal adjustment in static attention weights. This suggests the static stability of long-term relationships, and the model can learn stable attention weights.

3.5 System Observation

In this section, we introduce the observation process of the real-world trading system. Our model is trained and update strategy network in month cycles, with each training session utilizing data from the past two years as the training set. Firstly, our model undergoes industrial-level backtesting experiments in the testing system to ensure its adaptability to real markets, with a testing dataset spanning the past four years. The data acquisition system provides historical data and dynamically updated stock pools to the model, ensuring that no future information is introduced during testing. The investment portfolios generated by the model are processed by the strategy system to formulate specific investment strategies, which are then evaluated by the assessment system to assess the model's backtesting performance.

The industrial system utilizes the CSI1000 as the stock pool, from which we select two stocks with diametrically opposite trends, namely 600508.SH and 688639.SH as an example. We chose the CSI1000 dataset as it comprises 1000 companies from diverse industries and scales within the Chinese market. This provides a degree of market representation, enhancing the overall representativeness of our research. Figure 4 displays the investment interface where we record some contrasting investment trading signals for these two stocks in the system backtesting. The signal S represents sell mark, which is generated when the investment ratio of the stock on the current day is lower than the previous day. Conversely, the signal B represents buy mark and is generated when the investment ratio increases. For instance, on 2022-08-29, DGDRL's investment signals decided to reduce the holding proportion of 600508.SH and increase the proportion of 688639.SH. Results indicate the anticipated stock price decrease for 600508.SH and an increase for 688639.SH. The excellent performance demonstrates that DGDRL is applicable to real-market trading scenarios.

Following the system backtesting, we conducted a six-month real-time observational evaluation in the trading system to assess the practical profitability of DGDRL. During trading, the data acquisition system update the factor library and stock relation graphs in the local database the night before the trading day. Subsequently, it retrieves data within the look-back window and feed it into the trading signal system to generate investment portfolio signals using the pre-trained DGDRL model, which are subsequently made available for the trading system. After the first 15 minutes of the trading day, the trading system integrates these DGDRL investment portfolio signals. It compares these signals with the existing portfolio holdings, executing buy or sell orders for stocks that differ from the current portfolio to rebalance it. The evaluation results are depicted in Figure 5, where the first section displays the yield curves. The red curve represents the model's absolute returns, which correspond to the actual returns of DGDRL. The blue curve corresponds to the CSI1000 index, representing the market's overall returns. The yellow curve represents the excess returns, signifying

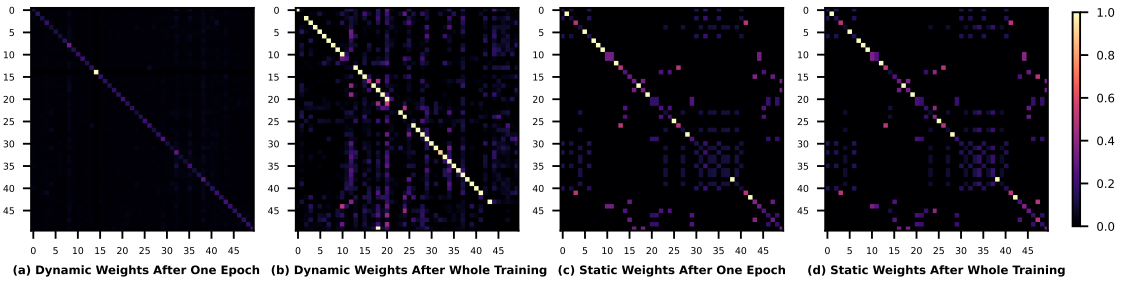


Figure 6: Visualization of dynamic and static attention weights. (a) and (b) display the dynamic attention weights, while (c) and (d) illustrate the static attention weights. We compare the results after the whole training and after just one epoch of SSE50.

the additional returns achieved by DGDRL compared to the market index. It is evident that DGDRL consistently outperforms the market, with accumulated excess returns exceeding 20%. Even in a market downturn scenario, DGDRL manages to limit losses below market, ultimately achieving positive returns. The second section provides the drawdown rate of excess returns, which signifies the magnitude of the decline in current excess returns from their peak to their lowest point. In other words, a lower drawdown rate indicates a smaller decrease in excess returns, reflecting the model's risk management capabilities. As observed in the figure, DGDRL maintains very low drawdown rates for an extended period, with occasional instances reaching around 2%. This demonstrates that DGDRL possesses the capability to pursue returns while prudently managing risks, achieving a balance between profitability and risk, making it suitable for real-world trading markets.

4 Related Works

4.1 Graphs in Stocks Relationship Modeling

Graphs are more accurate at representing relations between stocks than methods such as attention mechanisms because they can uncover hidden links of unrelated nodes [10]. Consequently, graphs have gained widespread adoption in quantitative investments [1, 9, 38]. For instance, Chen *et al.* [1] used historical features to represent asset nodes in graphs. Nevertheless, using temporal features alone to model stock relations is inaccurate and incomplete. Then news [4] and textual material [2, 24] are applied to graphs to better represent the changing relations between stocks. For example, CGM [45] incorporated financial news into heterogeneous graphs to represent relations. In order to make better use of graph information, graph attention [18, 32] is also employed to model relations between stocks [3, 17]. However, these prediction-based methods involve a two-step optimization process: profit prediction optimization and portfolio optimization. This inevitably leads to suboptimal performance as training target is temporal price trend rather than overall portfolio returns [25]. Our method can directly optimize the dynamic relation graph information end-to-end to increase the returns. It holds greater practical value and offers new insights for the utilization of graph neural networks in investment applications.

4.2 DRL in Portfolio Management

Compared with other methods, DRL can take maximization of investment returns as the training goal of the model, resulting in its widespread utilization in portfolio optimization nowadays. There have been numerous attempts, which combine the perceptual ability of DL with the decision-making ability of RL [5, 16]. AlphaStock [34] went a step further, constructing a risk-return balanced investment strategy with attention mechanism. Likewise, DeepTrader [35] then extended the framework by extracting macro market information. There are also DRL-based methods focused on modeling temporal price series [22, 39, 43], or mining for other factors[8, 11, 23, 40], and then extracting information to optimize portfolios. However, existing methods have not delved deeply into the exploration of stock relations with graphs, and the pioneers [29, 30] only defined relations between stocks as static. Our approach utilizes dynamic graphs to more accurately capture the temporal variations in stock relations, enabling us to make decisions that are most aligned with the current conditions. It provides valuable inspiration for further research on optimizing objectives by aggregating temporal dynamic graphs and features within the DRL framework.

5 Conclusions

In this paper, we proposed a dynamic graph-based deep reinforcement learning framework portfolio optimization for *naturally* modeling the long and short-term stock relations and *naturally* optimizing the portfolio. Firstly, we construct the stocks' relations into dynamic and static graphs. Then, we devise a multi-channel graph attention layer for efficient feature learning on dynamic graphs and infer the portfolio weights as actions directly, which are *naturally* optimized by portfolio rewards. Extensive experiments conducted on three real-world stock market datasets have demonstrated the superior performance of our proposed method. To the best of our knowledge, this is the first work that *naturally* optimizes portfolio decisions by incorporating the long and short-term stock relations in dynamic graphs. Our method has undergone empirical validation for its adaptability in real-world trading systems. It represents a departure from purely theoretical research and underscores its tangible and practical significance in the industrial context.

Acknowledgments

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